IOTAFLOW
SYSTEMS PVT. LTD

RPD SERIES
USER MANUAL

" If You Want To Manage It, Measure It "

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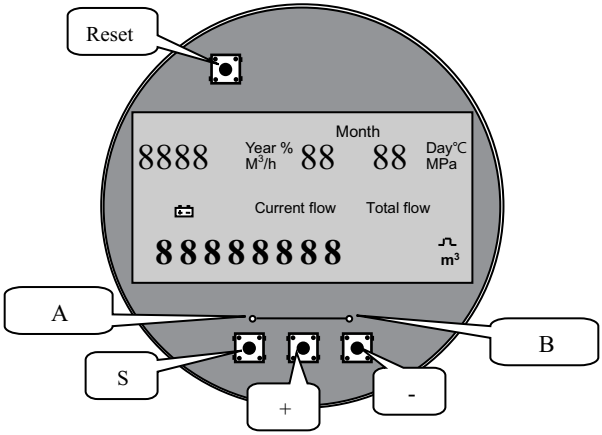
I. SUMMARIZE

Electronic Oil Meter - RPD Series is a battery operated, intelligent ultra-low power meter, The central processor adopts MSP430F147 chips, and piecewise compensation can be realized by setting the flow meter parameters, which can improve the flow meter's range ratio greatly.

2. FEATURE

(1) Using MSP430F147 chips, LCD display, ultra-low power consumption.
(2) Using iron-electric data storage, high reliable data protection function when power supply drop (high-speedstorage, data is not lost on countless times electric shock).
(3) Built-in clock chips, display time and date.
(4) High accuracy D/A conversion chip, guarantee the stability of 4-20 MA current output.
(5) Built-in Setting button and Reset button, and external has operation buttons, so that it can complete the full functional operation.
(6) The pulse output function.
(7) When the external environment temperature is below -5deg C, it can be heated automatically to guarantee the normal work of the screen.
(8) RS232 and RS485 communication inter face can work with computers and other intelligent instrument.

3. INSTRUMENT DIAGRAM



A and B arethe point of magnetic induction. When you switch display content (such as time), please use magnetic steel to across the above of A or B.

4. OPERATION

4.1 Meter para meter's code instructions

- **P00-P04:-** Flow meter's pulse equivalent(unit: L). When P22 is settled to 0, there is only P00 that's validity.

- **P05:-** Maximum flow (for 4-20MA, unit: m^3/h).

- **P06:-** Instantaneous flow has back to zero time, if it's more than this time, there's no signal and the time back to zero (unit: second).

- **P07:-** 4-20mA zerore-set (generally for 23, adjustable).

- **P08:-** To adjust full current (generally adjustable when 3950)

- **P09:-** To adjust the unit of instantaneous flow (m^3/h when setting as number 0 , L/min. When setting as number 1)

- **P10:-** Calculate circle of instantaneous flow (Unit : second)

- **P11-P14:-** Segmented value of flow meter ,unit : m^3/h (use the unit when P09=0)

- **P15-P18:-** Segmented value of flow meter ,unit : L/min. (use the unit when P09=1)

- **P19:-** Output pulse width . (Output high electrical level width, unit :ms)

- **P20:-** Pulse output model . (number 1 is timely pulse output ,number 0 I modulated pulse output)

- **P21:-** Pulse equivalent when adjusting pulse output .(valid when parameter P20 is 0)

- **P22:-** ON-OFF of segmented value. (unsegmented when number 0 and coefficient is P00 , segmented coefficient is P00-P04 when number 1)

- **P23:-** Correspondence Address . （ 0~16 ）

- **P24:-** Communication baud rate.(2400 when setting 0, 4800 when 1 ,9600 when 2)

- **P25:-** Indicating parameter. Indicating rotatively current flow and time when press button S and setting number 0 at the same time. Indicating rotatively totalized flow and time wen button S and setting number 1. Indicating rotatively current flow ,total flow and time when press button S and setting number 2.

- **P26:-** Return to original setting (Saving after setting 123 and exist)

4.2 Button Operation

(1). There are four buttons in total inside of flow meter . Upper one is button resetting ,those which are below of it are respectively s , + , - from left to right .

(2). Outside of it is a sensor button .

(3). When installing the flow meter and inserting the display board , press the reset button and then the screen starts to work .

(4). When at working situation ,press s and outside sensor button , the function is same .

(5). When at working situation , using magnetic steel to scratch outside of A 、B , the screen will shift to display daily flow , totalized flow and time .

4.3 Time setting

(1). Press s and + at the same time and then unstuck them , the screen will enter into time setting state and flickering the year .

(2). Press + or - and then adjust the year needed to set .

(3). Press s , flickering the month . Then adjust it .

(4). Adjust the date ,time as above rules .

(5). After adjusting the time , do as 4.4.4 instruction to save and exit setting program.

4.4 Coefficient setting

4.4.1 :- When at working or holding state , press s and + at the same time for twice, P00 flickering . The meter clock enter state of coefficient setting .
When state of time setting , press s and + at the same time for one time , the meter clock enter coefficient setting .

4.4.2 :- When parameter code PXX flickering , press + or - to find the corresponding code ,the below line will indicate the parameter which the code stand for.

4.4.3 :- If revising parameter , press s to move to the location needed to be revised , press + or - to plus or minus the value of parameter .

4.4.4 :- Save and then exit, press + and - at the same time, and then loosen the buttons at the same time, instrument showing -------- . Press + and - at the same time again for 3 seconds. If instrument shows End , parameters saving succeed, and setting end; if instruments shows Err , parameters saving fail, and need to reset.

e.g.：Adjust P00=000.12345，as the following steps：
I. Press s and + twice, entering to system setting.
II. Press s , the first digit flashing, and then press + and - , adjusting to 0.
III. Press s again, adjusting the second digit to 0.
V. Infer from this, adjusting the value of P00 to 000.12345, as the step 4.4.4, saving and exiting setting.

4.4.5 :- 4-20mA Zero set and full set

Wiring according to wiring diagram, connect with ampere meter, and get through external switching power supply.(12V or 24V)

(1). Zero setting: Entering to page P07 from parameters setting, press s , moving to parameters value, and then press + or - . When parameter's value changes, ampere meter's changes accordingly. The smaller the P07 value is, the smaller the current is. When ampere meter value is 4mA, press s , move to P07, flashing, and then press +, showing P08 parameters, zero setting end, and enter to full setting.

(2.) Full setting: parameters adjust to P08, steps as zero setting. By adjusting parameters, make ampere meter value to 20mA. When setting end, save as step 4.4.4 and then exit.

4.4.6 :- The method of flow segmented

Flow ratio defers from different flow meters. Especially when min flow rate gap is big, the measuring range will be influenced. But digital counter could overcome it by flow segmentation comparison. Instruction is as follows:

(1) When instantaneous flow adopt the unit of m3/h (P09=0), and actual instantaneous flow ≤P11, use P00 for pulse equivalent.

When P11<actual instantaneous flow≤P12, use P01 for pulse equivalent.

When P12<actual instantaneous flow≤P13, use P02 for pulse equivalent.

When P13<actual instantaneous flow≤P14, use P03 for pulse equivalent.

When actual instantaneous flow>P14, use P04 for pulse equivalent.

(2) When instantaneous flow adopt the unit of L/M(P09=1), and actual instantaneous flow ≤P15, use P00 for pulse equivalent; when actual flow ≥P18, use P04 for pulse equivalent.

(3) If without adopting flow segmentation, using one pulse equivalent, then just need to set P22 to. In this situation, P00 is pule equivalent for any flow rate.

4.4.7 :- Instrument adjusting method

(1) When adjusting instrument, set P22 to 0 firstly (no flow segmentation)

(2) Flow meter works at the middle flow, adjusting P00 pulse equivalent, making flow meter showing data equal to filling data.

(3) Choosing instantaneous flow unit, namely P09 is 1 or 0.

(4) Set P10 to be 1.

(5) Set P06 to be about 1/3 of the min flow. When instantaneous flow<P06, instantaneous flow shows 0.

(6) If without flow segmentation compensation, instrument adjusting ends; if with flow segmentation compensation, then follow step (7).
(7) According to the instantaneous flow of instrument, adjusting flow rate to min flow, calculate the equivalent by measuring tank and write it down.
Adjust the flow to different flow ranges, and respectively calculate the corresponding equivalent.
(8). Take as much as a few flow points throughout the flow range, and analyze the equivalent value in different flow points. To set the P11-P14 or P15-P18 according to the value of P09----0 or 1, then combine the different equivalent in P00-P04.
(9). Last, switch back and forth to different flow points to see whether the accuracy of the flow meter can meet the requirement.

5. CLEAR THE DAILY FLOW AND TOTAL FLOW

1. Clear the daily flow
Keep pressing the s key for 5 seconds till the daily flow on the screen displays 0, then loose the key to complete the procedure.

2. Clear the total flow
Keep pressing the - key for 10 seconds till the total flow on the screen displays 0, then loose the key to complete the procedure.
Remark:-
a. The measuring unit of daily flow is Liter, and the total flow is Cubic Meter.
b. The screen will display 4 decimal places after the clearance of daily flow and total flow.
c. When the numeric value on the screen is expired, the decimal point will automatically move back.
d. When no decimals is expired (that is 9,999,999), it will automatically do the clearance.

6. PARAMETERS

1. Input: pulse voltage
2. Sensor: a. Wiegand sensor(high frequency); b. A.H.Hall sensor(low frequency).
3. Display: use the Segment LCD.
4. Output: a. pulse voltage; b. 4~20mA
5. Communication interface: a. RS232; b. RS485
6. Communication protocol: Modbus
7. Power supply: DC3.6V, the battery lasts for 2 years, the battery icon will flash in low battery to prompt user to replace the battery.
8. Pulse output, 4~20mA output, communication and automatic heating require external power supply. External power supply shall be DC24V or DC12V.

7. WIRING

Terminal 1: Connect to DC24V or DC12V power+

Terminal 2: Connect to DC24V or DC12V power- (when two-wire system 4~20mA, 1,2 are not connected)

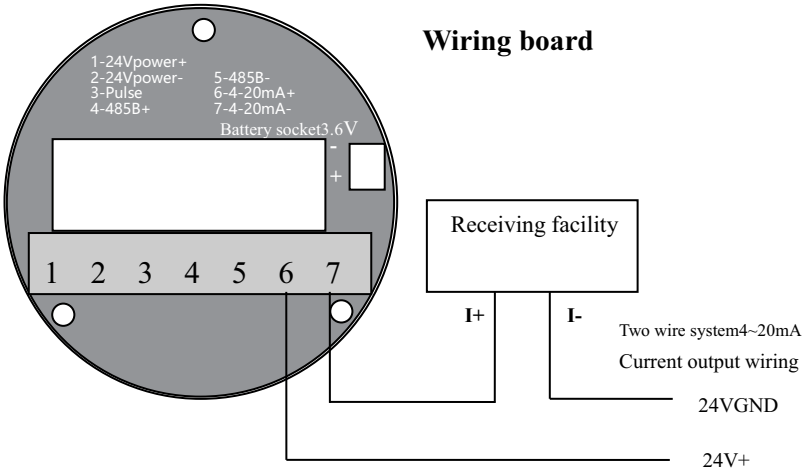
Terminal 3: Pulse voltage signal output

Terminal 4: Communication RS485A+

Terminal 5: Communication RS 485B-

Terminal 6: Two-wire system 4~20mA +

Terminal 7: Two-wire system4~20mA -



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